

THE COMPLETE DATA ANALYST ROADMAP

Go From Zero to Data Analyst in 6 Months

Learn with Bobby

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Welcome!

Hi! I'm **Bobby**, and I'm passionate about helping people break into tech—especially data-related careers. With years of experience in teaching and mentoring, I've created this guide to help **you** become a confident and job-ready **data analyst** in just 6 months.

This roadmap is structured to take you from a complete beginner to a proficient analyst step-by-step.

 Let's begin your transformation!

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TO LEARN THE APPTITUDE : <https://www.learnwithbobby.com/product-page/ultimate-aptitude-kit-1>

TO LEARN THE CODING: <https://www.learnwithbobby.com/product-page/coding-wizard-kit>

TO PREPARE FOR THE INTERVIEW: <https://www.learnwithbobby.com/product-page/interview-prep-kit>

Introduction

This roadmap will guide you through all the **core skills** you need to become a data analyst. Whether you're starting from scratch or transitioning from another field, this plan will help you build the knowledge, tools, and portfolio needed to succeed.

Target Audience

This guide is designed for:

- **Beginners** who want to become data analysts.
- **Intermediate learners** aiming to fill knowledge gaps or get job-ready.

Resources

To follow along, check out my free and premium content:

- **YouTube Channel:** <https://www.youtube.com/@withbobby>
- **Courses & Projects:** [LearnWithBobby.com](https://www.learnwithbobby.com)

Roadmap Overview

Skill	Est. Time	Phase
Mathematics & Statistics	1 months	Beginner
Excel	2 weeks	Beginner
SQL	2 weeks	Beginner
Python	1 months	Beginner
Version Control (Git)	1 week	Beginner
Data Collection & Prep	2 weeks	Intermediate
Data Visualization	2 weeks	Intermediate
Machine Learning (Optional)	2 weeks	Advanced
Big Data (Optional)	2 weeks	Advanced

 **Total Time: 6 months (with consistent daily practice)**

Mathematics & Statistics

Build a solid foundation to understand, explore, and interpret data.

 **Estimated Time: 1–2 months**

Key Topics:

- **Algebra:** Variables, equations
 - **Calculus:** Limits, differentiation basics
 - **Linear Algebra:** Vectors, matrices
 - **Probability:** Distributions, Bayes' Theorem
 - **Descriptive Stats:** Mean, median, standard deviation
 - **Inferential Stats:** Hypothesis testing, p-values, t-tests
 - **Regression Analysis:** Linear, logistic regression
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Excel

Excel is everywhere in data roles. Learn it well.

 **Estimated Time: 2–3 weeks**

Key Topics:

- Basics: Workbooks, formatting
- Formulas: SUM, AVERAGE, IF, VLOOKUP
- Charts: Bar, line, scatter, pie
- Pivot Tables: Group, filter, summarize
- Advanced Formulas: COUNTIF, IFERROR



SQL

Master querying structured data.



Estimated Time: 1–2 months

Key Topics:

- SELECT, INSERT, UPDATE, DELETE
 - Filtering, sorting, aggregations
 - JOINS: INNER, LEFT, RIGHT
 - GROUP BY, HAVING
 - Subqueries, Views, Transactions
 - Normalization, indexing, keys
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Python

Your primary programming language as a data analyst.



Estimated Time: 1–2 months

Key Topics:

- Basics: Variables, loops, functions
- Data Structures: Lists, sets, dictionaries
- File I/O: CSV, JSON
- Pandas & NumPy: Data analysis
- Visualization: Matplotlib, Seaborn
- OOP & modules (intermediate)

Version Control (Git)

Collaborate and track code changes effectively.

 **Estimated Time: 1–2 weeks**

Key Topics:

- Git Basics: init, clone, commit
 - Branching & Merging
 - GitHub: Pull requests, forks, code review
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Data Collection & Preparation

Clean, merge, and prep your data like a pro.

 **Estimated Time: 1–2 months**

Key Topics:

- Libraries: Pandas, NumPy, BeautifulSoup, Scrapy
 - Importing: CSV, Excel, JSON, APIs
 - Data Cleaning: Missing values, outliers
 - Data Transformation: Types, parsing, formatting
 - Data Integration: Merge, join multiple sources
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Data Visualization

Communicate insights through impactful visuals.

 **Estimated Time: 1–2 months**

Key Topics:

- Libraries: Matplotlib, Seaborn
- Tools: Power BI or Tableau
- Charts: Line, bar, heatmap, histogram
- Dashboards: Interactive and dynamic
- Data Storytelling: Narratives from insights

Machine Learning (Optional)

Go beyond analysis and build predictive models.

 **Estimated Time: 1–2 months**

Key Topics:

- Libraries: Scikit-learn
 - Supervised Learning: Regression, classification
 - Unsupervised: Clustering, dimensionality reduction
 - Evaluation: Confusion matrix, ROC, F1
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Big Data (Optional)

Handle massive datasets with modern tools.

 **Estimated Time: 1–2 months**

Key Topics:

- Frameworks: Hadoop, Spark
 - Storage: HDFS, NoSQL (MongoDB, Cassandra)
 - Processing: MapReduce, Spark Streaming
 - Ingestion: Kafka, Flume
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Final Words from Bobby

Becoming a data analyst is a **journey**, not a sprint. Stay curious. Keep experimenting. Build projects. Ask questions. And remember:

“Every expert was once a beginner.” — Bobby

Let's grow together: LearnWithBobby.com